

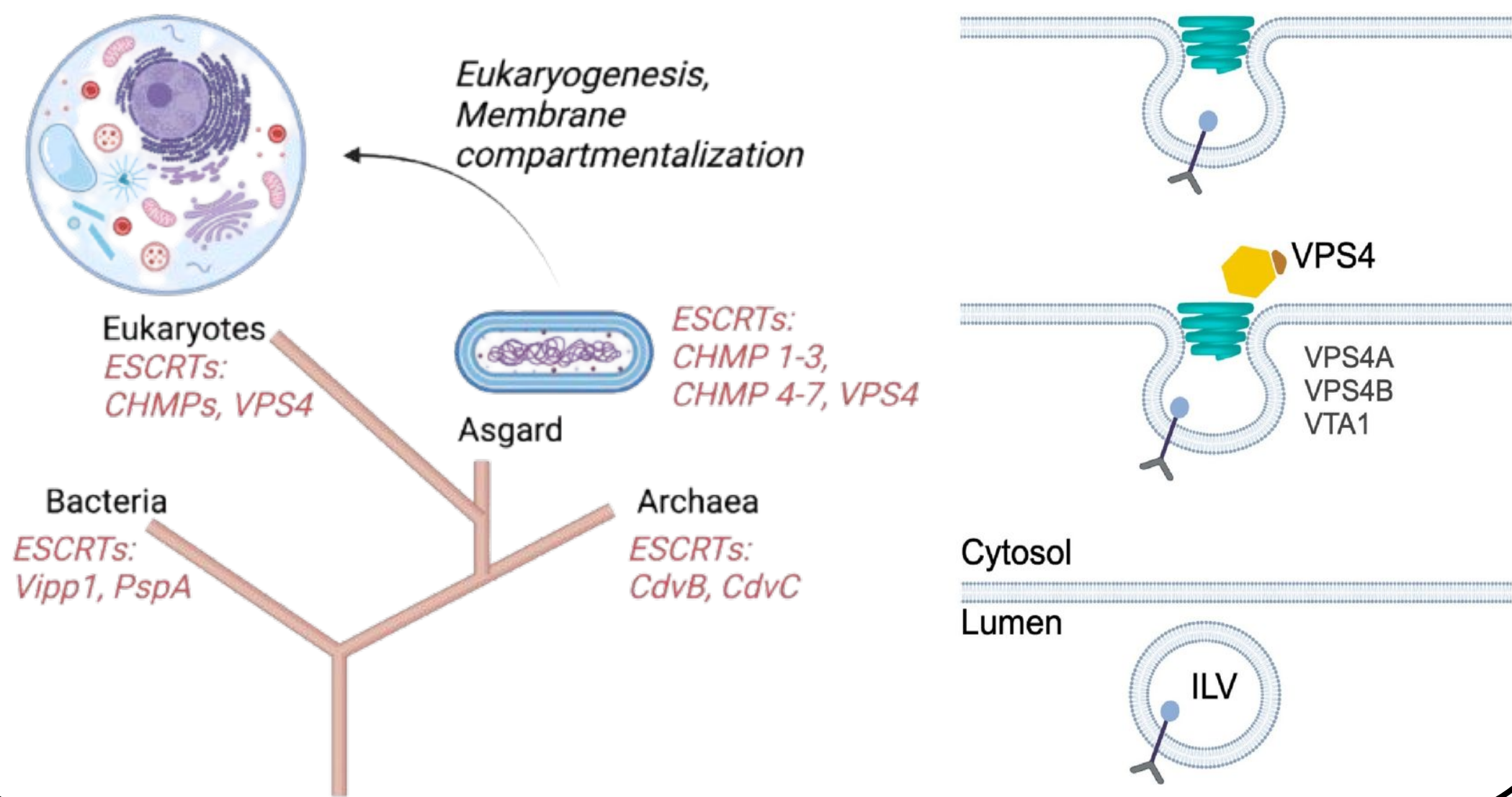
Environment drives residue-level adaptations while phylogeny dictates the biochemical backbone of ESCRT-III proteins in Asgard archaea



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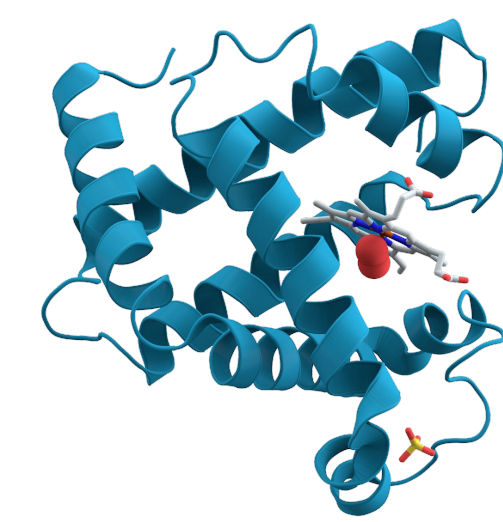
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Introduction

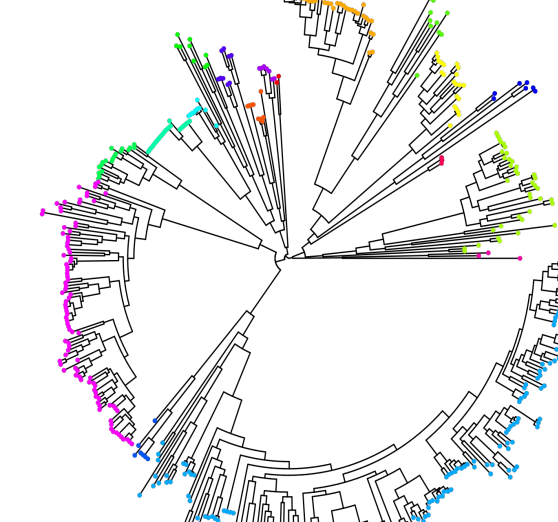


Our working hypothesis

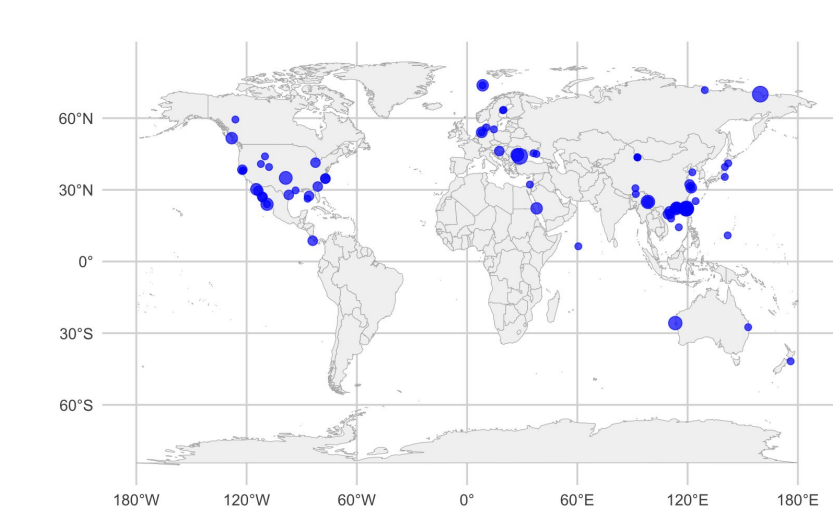
$$\text{Similarity} \sim \text{Phylogeny} + \text{Geography} + \text{Environment} + \epsilon$$



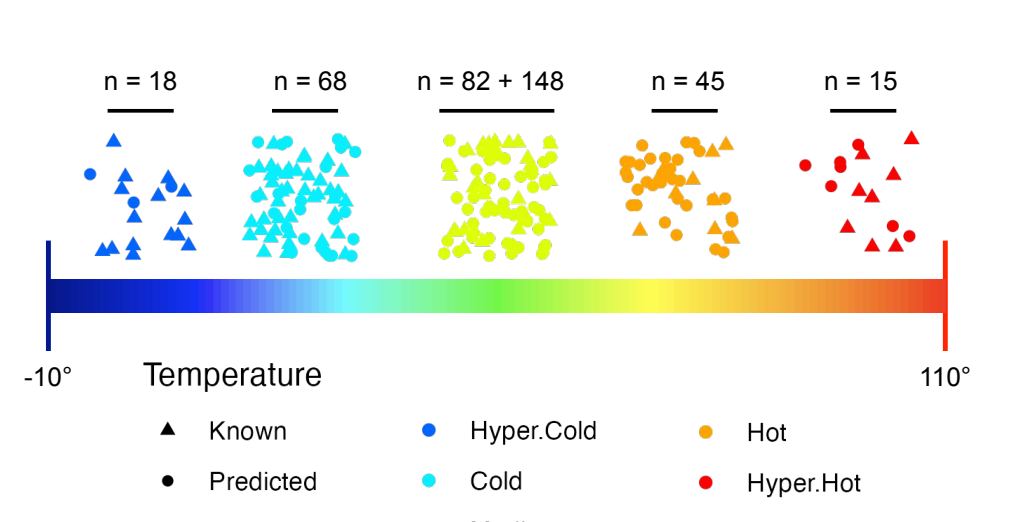
Sequence,
Structure



Mutations,
Evolution, ...

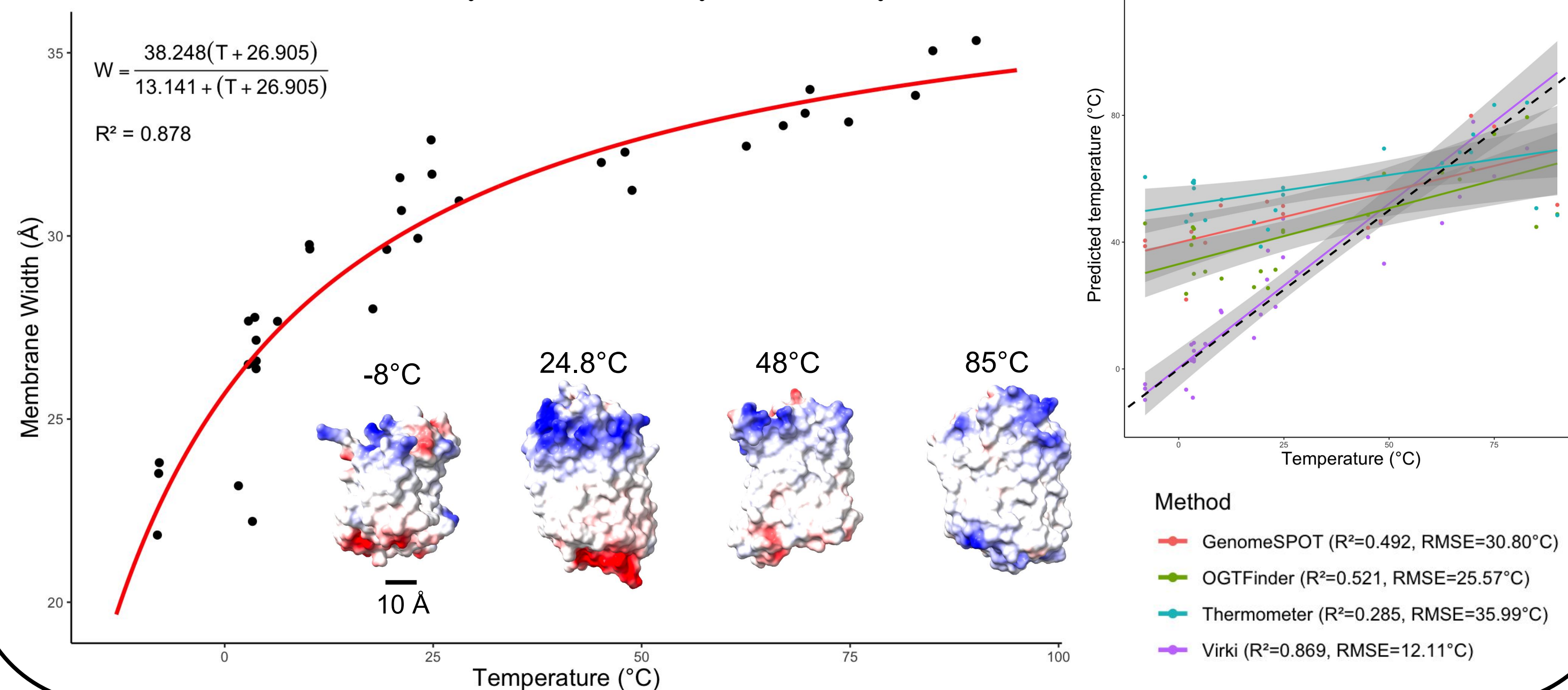


Movement,
HGT, ...



Temperature,
Salinity, ...

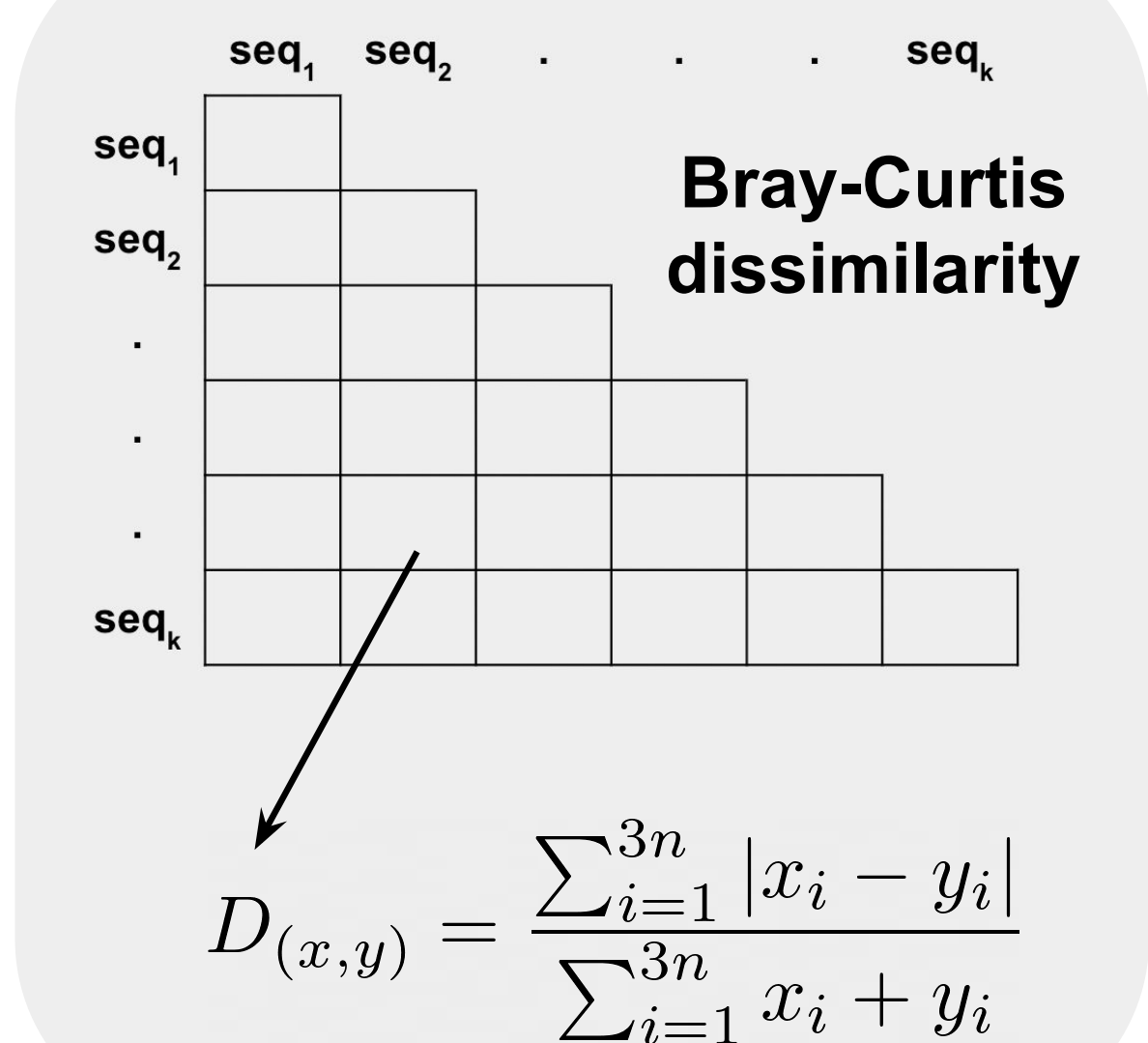
A method to complete sample temperatures



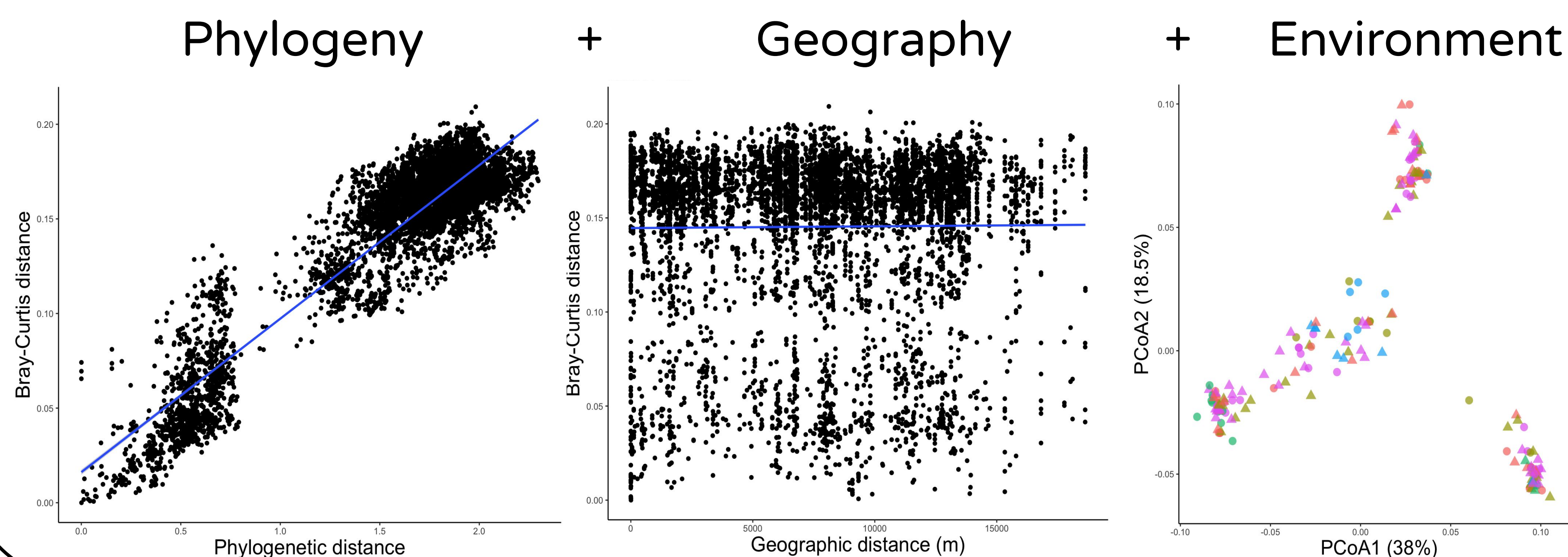
Measuring protein distance with Bray-Curtis

Seq₁: -----IKDEIAESELK-----
Seq₂: -----LTDELKEGDT-----
...
Seq_n: -----LQDEVEKKELKS-----

 $v_x = \dots (\text{Hyd}_{93}, \text{Polar}_{93}, \text{Bulk}_{93}), (\text{Hyd}_{94}, \text{Polar}_{94}, \text{Bulk}_{94}) \dots$



Phylogeny is the main driver of sequence similarity



Identifying residue-level changes

For each
residue

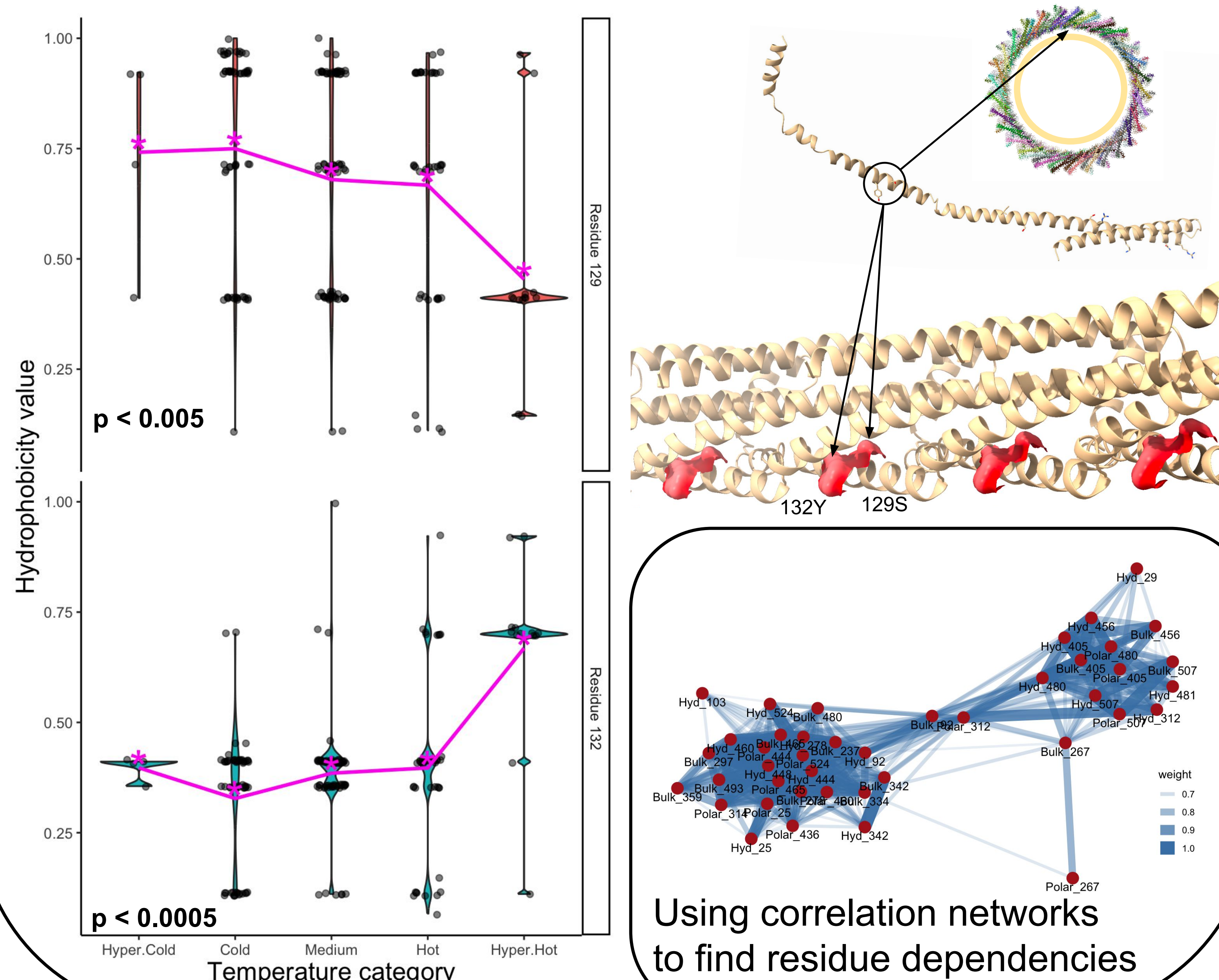
$v_1: (0.00, 0.57, 0.62, \dots, 0.46, 0.21)$
 $v_2: (0.69, 0.12, 0.29, \dots, 0.05, 0.48)$
...
 $v_n: (0.78, 0.56, 0.67, \dots, 0.73, 0.90)$

ANOVA
(p)
+
Pearson
(r)

Residue	p _{adj}	r
Hyd ₁₀₁	0.003	0.53
Polar ₂₉₁	0.192	0.04
...	...	...



Residue-level changes and interactions are induced by environmental conditions



Conclusions

- * We developed a methodology that utilizes the biochemical properties of the residues, unfolding changes not detected in alignment.
- * Phylogeny is the main driver of the biochemical backbone and geography shows no strong effect on the ESCRT-III proteins.
- * Environment induces residue-level changes with predicted functionality in the ESCRT-III proteins.

Bibliography

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2. Nachmias, D. et al. ESCRTs – a multi-purpose membrane remodeling device encoded in all life forms (2025).
3. Panagiotou, K., Geesink, P., Köstlbacher, S. et al. Diversity, ecology, cell biology and evolution of the Asgard archaea (2026).

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online poster at
our website!